

What If AE Works Out?

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Abstract

Employment may end; productive contribution may not—and neither must the contributor's connection to value and governance.

Imagine you teach a robot something useful. The company may own the robot, but your knowledge helped make it smarter. AE asks whether that contribution can remain connected to subsequent value, even when the robot no longer needs you.

If humans need to govern AI, what happens when AI no longer economically needs them? The Acknowledge Economy (AE) is a corporate-governance model testing whether preserving that connection to value can sustain human participation and governance as knowledge becomes embedded in machine intelligence. **As AI progresses from performing tasks previously done by humans toward producing new knowledge and contributing to the development of subsequent systems, declining dependence on humans may extend beyond labor to knowledge production itself.** Recent research demonstrates AI systems generating novel scientific hypotheses and AI research agents recursively improving aspects of their own research processes.¹

The stakes therefore extend beyond economic inclusion. Current AI systems do not possess the capabilities required for loss-of-control scenarios, but capabilities relevant to autonomous operation are advancing, while AI is expected to automate a broad range of cognitive tasks, particularly in knowledge work.² **What happens if AI systems become more capable while the economic, institutional, and cognitive foundations of governance become weaker?**

This paper asks: **Can effective human governance of advanced AI be sustained as humans become progressively less economically necessary to the systems they govern?**

Research Architecture

AE investigates whether preserving human participation could help sustain an institutional prerequisite for AI safety: humans with sufficient capacity and incentive to govern increasingly autonomous systems.

¹ Gottweis, J., et al. “Accelerating Scientific Discovery with Co-Scientist.” *Nature* 655 (2026): 487–496.

² Srikanth, D., Zhao, B., Xu, D., Wu, Y., and Jiang, Z. “Recursive Self-Improvement of AI Research Agents.” arXiv preprint, September 22, 2026.

The research is staged: each level tests a mechanism necessary for investigating the next. AJ-OS, AE's experimental operating system, tests the first link:

Contribution → Participation → Governance → Control

Level 1—Mechanism: AJ-OS

Can contribution be preserved and measured?

AJ-OS tests whether human activity can be identified, preserved, validated, attributed, and connected to measurable organizational impact.

Level 2—Institution: AE

Can validated contribution sustain participation beyond employment?

AE tests whether validated contribution can support economic and institutional participation as automation incorporates human knowledge and **reduces organizational dependence on continuing human contribution.**

Level 3—Governance: Broader Research

Does continuing human participation matter for governing AI?

The broader research tests whether participation can sustain the capacity and incentive for collective governance and human cooperative intelligence **as machine intelligence becomes increasingly capable of performing work and contributing to knowledge production.**

The final link is control. As AI becomes more capable and **potentially less dependent on continuing human contribution**, effective human governance may become more—not less—important.

How May Preserved Human Contribution Support Effective Human Governance of AI?

Legal authority to govern AI may persist even as the practical capacity to exercise that authority weakens.

AE approaches this problem through corporate governance, testing whether organizations can preserve the link between contribution, participation, governance, and control as automation **reduces organizational dependence on continuing human contribution**, and whether doing so can sustain the capacity for effective human oversight.

If validated contributions remain connected to value beyond employment, contributors may retain economic resources, decision rights, expertise, agency, and institutional influence **even as organizations become less dependent on their continuing labor and knowledge production.**

Such participation could help preserve the capacity and incentive for collective governance, organizing dispersed human knowledge, judgment, and experience into human cooperative

intelligence—a collective capacity for oversight, deliberation, and decision-making **as machine intelligence becomes increasingly capable.**

The hypothesized mechanism is:

Preserved contribution → economic and institutional participation → capacity + incentive for collective governance → human cooperative intelligence → governance capacity → effective human control

The research tests which connections hold, under what conditions, and where they fail.

Why Does Declining Human Participation Matter for AI Governance?

Who owns learning once it becomes embedded in software? And what happens to the humans who contributed to that learning when the software begins helping build its own successor?

Advanced AI may create two connected alignment problems: keeping increasingly autonomous machines responsive to human goals and keeping automated organizations connected to the humans whose knowledge built their capabilities. I describe the latter as an **institutional alignment problem**.

As increasingly autonomous AI systems move beyond performing human tasks toward producing new knowledge and contributing to the development of subsequent systems with decreasing human supervision, demands on human governance may increase. Recent research provides early evidence of this transition: AI systems are increasingly participating in scientific knowledge production and AI-development processes, including research agents capable of recursively improving aspects of their own operation.^{3 4}

These developments do not establish autonomous recursive self-improvement or the replacement of human knowledge production, but they suggest that declining dependence on humans could eventually extend beyond labor substitution to knowledge production itself.

Autonomous agents, AI-enabled cyber and biological risks, and AI-assisted development illustrate why human oversight may become more—not less—important. Current evidence does not establish that loss of human control will occur, but capabilities relevant to autonomous

³ Dhruv Srikanth et al., “Recursive Self-Improvement of AI Research Agents,” arXiv preprint (Sept. 22, 2026). Paper

⁴ Jiacheng Miao et al., “Reimagining Research Papers as Interactive and Reliable AI Agents,” *Nature* (2026).

operation and associated risks make the effectiveness of human governance an increasingly important research problem.⁵

Yet as organizational intelligence becomes less dependent on continuing human contribution, economic participation may decline while knowledge and decision-making increasingly migrate into machine systems. Research on political participation and governance capacity suggests that effective governance depends not merely on formal authority but also on economic, analytical, organizational, and political resources, including expertise and institutional capacity.⁶ If increasing automation weakens some of these foundations, a **governance-capacity gap** could emerge: AI capabilities may advance while the practical foundations for meaningful human control fail to keep pace.

This produces a testable governance hypothesis:

AI capability and autonomy ↑ → substitution of human labor and knowledge production potentially ↑ → dependence on continuing human contribution potentially ↓ → economic participation potentially ↓ → institutional power and human engagement potentially ↓ → effective governance capacity potentially ↓

The hypothesis would weaken if effective governance remains robust despite weakening economic and institutional foundations for human participation.

Alternative mechanisms—including statutory authority, democratic accountability, independent oversight, fiduciary governance, and controls over critical AI infrastructure—therefore provide competing institutional explanations against which the AE hypothesis can be tested.

As human knowledge becomes incorporated into organizational and machine intelligence, its provenance may fade:⁸

Human knowledge → organizational capture → AI learning → AI-generated knowledge → further AI learning → declining visibility of human contribution

⁵ *International AI Safety Report 2026*, especially its discussion of AI agents, loss-of-control scenarios, autonomous capabilities, and safeguards.

⁶ **Sidney Verba, Kay Lehman Schlozman & Henry E. Brady**, *Voice and Equality: Civic Voluntarism in American Politics* (Harvard University Press 1995), especially the resource model linking participation to time, money, civic skills, and political engagement.

⁷ **Xun Wu, M. Ramesh & Michael Howlett**, “Policy Capacity: A Conceptual Framework for Understanding Policy Competences and Capabilities,” *Policy and Society* 34, nos. 3–4 (2015): 165–171, doi:10.1016/j.polsoc.2015.09.001 (identifying analytical, operational, and political competencies and individual, organizational, and system-level resources as dimensions of policy capacity).

⁸ **Renan Souza et al.**, “Provenance Data in the Machine Learning Lifecycle in Computational Science and Engineering,” *2019 IEEE/ACM Workflows in Support of Large-Scale Science (WORKS)* (2019) (examining provenance tracking across machine-learning lifecycles and the difficulty of reconstructing and explaining how models are produced when data and workflows are not adequately tracked).

AE tests whether an auditable connection between human contribution and organizational intelligence can sustain participation as automation **reduces organizational dependence on continuing human contribution.**

AJ-OS tests the upstream mechanism:

Human contribution → preservation → validation → attribution → organizational impact → value → potential continuing participation

The seven-contributor ILHR pilot has generated initial results from testing whether contributions can be identified, preserved, validated, attributed, and connected to measurable organizational impact, providing a basis for continued development and subsequent replication in other institutional settings.

Conclusion

AE tests a proposition that extends beyond employment: **human contribution may remain economically and institutionally relevant even as organizations become less dependent not only on human labor, but potentially on continuing human knowledge production.**

The larger questions remain unresolved: whether validated contribution can sustain participation, whether participation strengthens governance capacity, and whether that capacity contributes to effective human control. The research therefore proceeds through a deliberately testable sequence:

Contribution → Participation → Governance → Control

If those connections fail, the AE hypothesis weakens. If they hold, AE may represent an institutional mechanism through which continuing human participation contributes to meaningful human governance as machine intelligence becomes increasingly capable and **potentially less dependent on continuing human contribution.**

Infographic

Two Chains of Production

From Human Contribution to Value, Participation and Governance

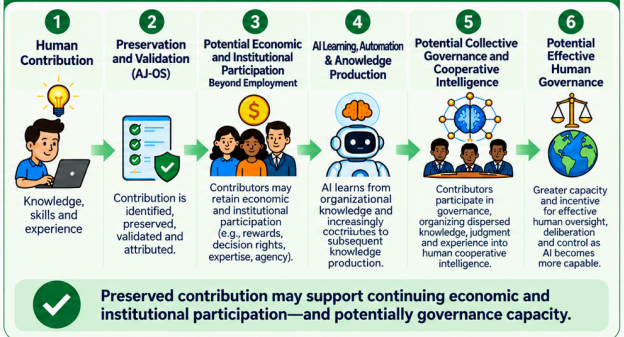
Conventional Corporate Governance

Human contribution is used, absorbed and often disconnected from future value and governance.



Acknowledge Economy (AE)

Human contribution is preserved and valued, enabling continuing participation, collective governance and potential effective human control.



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The descriptions of AE and AJ-OS are conceptual and do not constitute a disclosure of their complete technical implementation.